

The Ceiba tree: An expansive blossoming of dramaturgical structures for audiovisual expression.

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Abstract

The ceiba tree, also known as the kapok tree, is a generous species found in tropical forests around the world. It stores water in its trunk to prepare for droughts, which benefits itself and other trees. Additionally, the ceiba has the potential to nurture the transdisciplinary, two-way relationship between humans and the plant kingdom. This broader perspective sheds light on new ways to expand the limits of traditional dramaturgy and audiovisual language by linking plant life and alternative plot structures. Cultivating the audiovisual human-plant interaction (HPI) is possible thanks to critical perspectives on prevailing anthropocentrism and specialization, such as post-humanism and ecofeminism, where empathy and care practices are in the center. This paper presents a transdisciplinary, more-than-human dramaturgical framework comprising four steps, which focuses on co-creation with plant life. Our blueprint demonstrates that more-than-human dramaturgy involves relativizing human structures and developing interspecies expression systems.

1. Introduction

Given the anthropocentric character of the world in which we live, it is unsurprising that current dramaturgy strategies employed by the audiovisual and theater industries focus primarily on stories about humans, made by humans, for humans. In the context of global warming and the profit-based relationship we have with the environment, as described by authors such as James Bridle [Bri22], it becomes urgent to find new ways of relating with the more-than-human. This is not only because we depend on them, but also because they are equally important as humans and must be given the same level of respect and dignity. This paper focuses on the plant kingdom, which has been pushed into the background, both literally and figuratively. Considering that the way we create and experience the stories is constantly shaping and being shaped by the life outside the cinemas, theaters and screens, it is crucial to develop a more-than-human dramaturgy that emphasizes the importance of the plant kingdom as an active subject of our society. This will enable us to create audiovisual experiences that impact the way we behave and interact with the more-than-human beyond the artworks. Failing to do so will perpetuate the existing hierarchical, profit-driven relationship with the More-than-human and reinforce the division between humans and the plant kingdom.

The four-step framework presented in this paper is a tool for learning, creating and telling stories in multifaceted and creative ways in relation to the plant kingdom. This venture takes a transdisciplinary approach, considering storytelling not only as an artistic domain, but also as a cross-cutting issue in all disciplines. As

plants and trees communicate in ways that are not human-centric, the bridge we construct in this paper is a dynamic, evolving structure that allows the plant kingdom to participate in audiovisual creation as a subject, rather than merely as a prop.

Scientists, philosophers, artists and dramaturges have independently undertaken the challenging endeavor of developing a more-than-human perspective. This paper presents a framework that combines and solidifies these perspectives into a creative method for projects integrating science, technology, and art. The ceiba tree inspired this framework. Its generosity and interconnectedness with the tropical forest are key to creating a dramaturgy in which different beings support each other and are all equally important to the ecosystem. To our knowledge, there has been no general dramaturgy framework focused on the plant kingdom, let alone one with a more-than-human perspective. This is therefore the paper's main contribution.

After reviewing the related work, we describe each of the four parts of the framework. Each part is a step of the path that we propose for co-creating audiovisual experiences with the plant kingdom: Contextualization, Recollection, Storage/Transformation and Sharing/Building. The following sections discuss the conclusion, challenges and future work. The framework's transdisciplinary approach can provide answers that a specialized approach cannot. At the same time, it can prompt new questions about the future of the more-than-human with the human, considering how the audiovisual industry has evolved in recent years and how it can evolve next. The plant kingdom is already telling a story, and we can learn from

it and expand the limits of our human perception and storytelling patterns. In doing so, we can create new audiovisual expressions in which plants and trees play an important role in the creation of artworks and build a present and a future in which they are essential members of society beyond the screen and art institutions.

2. Related Work

In order to create a dramaturgy framework with a transdisciplinary and more-than-human approach where the plant kingdom is placed at the center is important to talk about fundamental works that have shed some light on the topic. One starting point was the interdisciplinary framework offered by the work of five women scientists Chang et al. [CM22], to design the Human–Plant Interaction (HPI) in accordance with the Human–Computer Interaction (HCI). This literature review focuses specifically on living plants, where science, art, sociology, sustainability and a more-than-human perspective coexist. Their work focuses on artifact construction and implementation, which is very useful. However, they say little about audiovisual or dramaturgical possibilities, since their artistic approach is architectonic and design-based. Our framework is complementary and can certainly broaden the transdisciplinary more-than-human perspective by demonstrating that audiovisual media, dramaturgy and scientific technological advances can coexist harmoniously.

Another relevant work was created by Brandstetter [Bra24]. In it, the author discusses the artistic practices of Ruth Geiersberger and describes how art can transform the way we think, feel, and relate to plants. The author also proposes dance and performance as a means of combating colonial and anthropocentric practices. The use of prepositions: "with", "for", "about", "as" can definitely be part of the way in which we question and rebuild our relationship with plants through language and performative approaches. It is not the same to tell a story with plants, for plants, about plants or as plants. This exploration already brings us closer to the audiovisual expression, since Ruth Geiersberger has recorded her performances and offers the language as a tool. Our framework extends beyond this, as we examine classical and alternative dramaturgical structures of the narratives we can recount, thereby augmenting the reflection on prepositions of this inspiring work.

Žeželj [Že22] discusses more-than-human dramaturgy and provides a theoretical framework based on feminist discourse on water by Mielle Chandler and Astrida Neimanis, as well as Donna Haraway's reflections. Complementing the two previously described related works, this one conveys a radical and significant message: we do not need to comprehend or know the more-than-human to coexist with them. Agency should not be a prerequisite for the consideration of other beings in politics and ethics; therefore, empathy, responsivity, and sociality become crucial. Our framework is also based on the intersection of feminist environmental humanities and artistic expression. We also see dramaturgy as a collaborative process between different bodies. However, in contrast to Žeželj's work, we consider classical dramaturgy theorists and alternative plot structures in our analysis. While our framework considers plants and trees to be the theme, characters, and co-creators, the author analyzes art practices involving ghosts and the moon.

Eg-Tverbakk [ET23] has written about the dramaturgy as a

collective and dialogical practice. She calls it "Dramaturgy of assemblages" and focuses on new materialism. In her work, the dramaturg is a nomadic being who, rather than interpreting and structuring the text, feels, captures, and discovers the structure in the intermittent rhythms, energies, and atmosphere of all the things and subjects involved in the creative process. This approach requires stepping back and letting go of the ego and the need to give form to the artwork. Like our framework, her work is based on posthuman thinking and collaborating with the more-than-human. However, while she describes an attitude toward the creative process, we complement her perspective by taking into account the classical and alternative plot structures used in film creation, as well as technological and scientific contributions. These aspects can enrich the creation of theater and audiovisual artworks without losing the author's sensibility and keeping transdisciplinary collaboration alive. Furthermore, while she speaks about all bodies and things involved in the creative process, we chose to focus on the plant kingdom.

3. The dramaturgical plant life framework

Similar to how soil, water, and sunlight contribute to a plant's growth, the framework we propose emerges from a fertile foundation of existing approaches to interaction with the plant kingdom and dramaturgical possibilities. Traditionally, the creation of human knowledge has been divided. Our goal is to offer a conjunction of science, art, philosophy and dramaturgy that can serve as a tool for audiovisual creators who are interested in respectfully and curiously working with the plant kingdom and who are eager to learn from and be shaped by other beings, instead of exploiting others for their own selfish benefit.

In mainstream western culture, the relationship between humans and plants has been hierarchical. Even though humans depend on plants and trees for fundamental things like breathing and eating, they are seen as an exploitable, profitable resource, disregarding the fact that they are living beings that guarantee the balance of the ecosystem, that their needs are equally important to those that make human life possible, and that they need our support and care. Our interactions with the plant kingdom and the stories we tell reinforce the idea that we have needs and make decisions, while plants are passive objects. However, in light of climate change and social inequality, it is clear that we must reevaluate our connection with the plant kingdom so that all species, including our own, can survive. This requires a change in the way we create discourse about humanity and our relationships with other beings.

In order to do so we propose a framework based on the ceiba tree. Upon observing this species, two parallel processes with different temporalities were identified, inspiring the four-step blueprint presented in this paper. On the one hand, this tree, like other members of the plant kingdom, transforms carbon dioxide, which is largely produced by humans, into oxygen in a constant process of exchange for the production of its own food, due to its autotrophic nature. This is a good time to remember that it is thanks to this kingdom that the planet's oxygen reserves, and therefore our lives, are guaranteed. On the other hand, this tree is able to store water in its trunk during the rainy season, in anticipation of the dry season. The pace of the process is slower in comparison to photosynthesis. The impressive behavior of this being is that when the drought

comes it shares its water with all other surrounding trees regardless of whether they are of another species. We believe that the generous behavior of this species is in the core of a more-than-human dramaturgy where plant life and humans tell and construct together.

Based on these biological facts we describe the 4 steps of our blueprint. Firstly, we see the constant relationship between the artwork and society as analogous to the way the ceiba and the environment feed back into each other constantly in photosynthesis. The first step of our framework, called contextualization, sheds light on the nature of this relationship. Secondly, we propose that the way in which we conceive and visualize the creative process of a more-than-human artwork in which a storytelling process with plants can exist is similar to the way in which the ceiba generously collects, stores and shares water with other species. Steps 2, 3 and 4, called recollection, storage, transformation and sharing-building, elaborate on the details.

3.1. Step 1: Contextualization

The favorable environment in which plant life can flourish is as important as plant life itself. Together, they form a symbiotic relationship in which both shape and are shaped by each other constantly, as is the case with photosynthesis. The preliminary stage of a more-than-human dramaturgy, in which a co-creation process with the plant kingdom is possible, are concepts, authors and theories that have questioned the anthropocentric, extractive, colonial and chauvinistic practices that have permeated the foundations of our society and which have placed greater importance on humanity than on anything else.

It is important to be familiar with posthumanism, ecofeminism, alternative artistic and dramaturgic and empathy in order to co-create with the plant life. The existing division between humans and plant life mirrors the division between “civilized human beings” and “the others” in colonial processes such as the invasion of America during the 15th and 16th centuries. These divisions and hierarchies led to humanistic approaches by persons who spoke out against this unfounded categorization. Montaigne [Mon16] and Tzvetan Todorov [Tod10], for example, vindicated indigenous societies and other groups that were enslaved and colonized by defending their ways of thinking, feeling and creating, which are characterized by placing care for nature and diversity at the center.

Posthumanism and ecofeminism elaborate on that vindication by including more-than-human beings. We recommend the work of Forlano [For18], who offers a critical approach to different types of posthumanism of different relevant authors, leaving us with a relevant reflection about the importance of questioning the relationship between humans and non-humans. Moreover, the author questions whether talking about posthumanism means that humanism has already been solved. There are still challenges in recognizing the humanity of woman, black communities, Indigenous peoples, LGBTQ+ individuals, and those who suffer from xenophobia and classism around the globe. Focusing on plants does not mean stopping the pursuit of a better world for humans. Forlano’s analysis includes key authors whose discourses are necessary to understand posthumanism and which are essential for the creation of more-than-human approaches.

Art is an institution that has generated hierarchies, which have reinforced dominant structures. However, some authors criticize that and have shared perspectives compatible with posthumanism and ecofeminism are key to co-creation with plants. Gene Youngblood [You70] spoke about the expansion of cinema and how art can reunite what humans have separated. He redeems the importance of audiovisual explorations beyond the canons, by asserting that they allow consciousness to expand when metaphysics are vindicated in a rational, specialized, unequal world lacking environmental or universal consciousness. He affirms that those views are what lead to the unequal accumulation of capital, the disconnection between cultures, and the unsustainable exploitation of natural resources. Youngblood calls those explorations “Synaesthetic cinema” and defines them as an audiovisual media that goes beyond commercial cinema and is the only aesthetic language appropriate for the intermedia world in which we live, as it understands the non-linear, non-uniform, and unconnected nature of the world we live in, basing itself on it, rather than creating order out of chaos, as narrative cinema does. Synaesthetic cinema goes beyond turning points, drama, and story and allow us to be aware of and express the phenomenon of our own consciousness and perception of the human condition. Youngblood affirms that computers, television and new media are an extension of our central nervous system and we can use that to our advantage in the expansion of art.

The industry of entertainment profits from the maintenance of plot, story, and drama by producing these effects, satisfying these preconceived expectations. Matei Calinescu [Cal03] describes the consumer of that industry as someone who wants to fill his free time with maximum excitement derived from culture, but without effort. It is a person who wants to be beautifully lied to, is mentally passive, and spiritually lazy. Umberto Eco [Eco13] affirms that the industry of entertainment preserves the canon, while art is a system of systems in which there are internal relationships, with the audience, and with the historical context. Art is interested in the processes that lead to the artwork, the form, and makes use of a poetic message, which consists of a model of communication characterized by ambiguity rather than univocity. To this end, it eliminates redundancy and therefore leads the recipient of the message to decode the message by deducing a code unknown to them, using not the information in the message but the context of the message, their intelligence, and sensitivity. This causes the receiver to be surprised and engage with the message, focusing their attention on the structure of the message and its multiple meanings. This can be an adventure for the audience, as this effort to discover a new code leads the audience to converge on all kinds of hypotheses that stimulate their psychological and intellectual disposition.

Youngblood, Calinescu and Eco describe the level of superficiality that can have the entertainment and the challenge we have in order to expand the panorama of the art, the audiovisual and of course the dramaturgy. The synaesthetic cinema and the poetic message described by Youngblood and Eco are a bridge between the posthumanism, the ecofeminism and the more-than-human dramaturgy.

Le Guin [Gui19] criticizes the way that the story of a hero who is hunting has been preferred and seen as the story of humanity. She invites us to look for stories of life instead of continuing to tell stories of killing. She mentions the possibility of recognizing our-

selves as humans in the act of containing energy instead of extracting it and in the act of caring instead of killing. Humans have not only been hunters, but gatherers as well. In other words, Le Guin has not only identified the dominant story, which has been used not only in art but also in entertainment, but also suggests taking a different approach to storytelling, in which a more-than-human discourse is possible.

This resonates with the way in which McKee [McK97] explains the plot. McKee affirms that all stories have a plot, the difference lies between Archplot, Miniplot, Antipplot and Nonplot. Archplot is the classical story of the mainstream films of a single protagonist who wants something and will do anything to get it while the external circumstances (conflict) work against that desire in a cause-effect chain in a continuous time frame, while building up an increasing suspense arc until a closed ending answers all the questions and satisfies all emotions. In Le Guin terms, the Archplot would be the story of the hunter. According to McKee this type films express the interconnectedness of reality. Archplot, says McKee is not western or Eastern but human. To reduce the plant kingdom to a character of one of those films would still force the human perspective on the plant kingdom, because it is telling the same story and not expanding the possibilities of the dramaturgy.

Miniplot is a minimalist approach in which the Archplot is reduced. There is an open ending, an internal conflict and multiple reactive or passive protagonists. However, the audience could be able to answer the open questions and resolve the unfulfilled emotions. Antipplot is the cinema version of the antinovel and the theatre of the absurd. There are inconsistent realities, nonlinear fragmented time and coincidences instead of causality. McKee says that this kind of films tend toward extravagance and deliberate exaggeration and leads to absurdity or nonsense, since the disconnectedness of the existence is expressed. The Nonplot refers to films comparable to a portraiture, where the initial and final state of the character is the same so no story is told, but the audience has an insight and is transformed in the best-case scenario.

However, there is another aspect that is important in the creation of a more-than-human dramaturgy framework: the consistency of the fictional world's rules. For McKee the consistency of the reality is the maintenance of the rules that govern the modes of interaction between characters and the fictional world of the film. The Archplot and the Miniplot have consistency, but the Antipplot is an inconsistent reality, in which the rules are made to be broken so the modes of interaction change from one reality to another, creating the sense of absurdity, from a human point of view of course. According to the author, the Antipplot can be the expression of the subjective state of mind of the creator, in which there is not sane or insane but symbolic and ideational.

To go beyond the human means to get closer to the unknown and the Miniplot with its open endings, the Antipplot with its unique ways of expressing the creator's perspective and the Nonplot with its portraiture quality, can give the plant kingdom the tools to express themselves. The way in which we tell stories is still mainly anthropocentric. If we want to build a more-than-human dramaturgy it is imperative that we question the traditional categories. The plant kingdom works as a network with many nodes whose behavior is most similar to the Antipplot or Nonplot of McKee, because the way

in which we relate to those beings is undermining their needs and the interconnectedness that exist between them and us. Only because we cannot comprehend them completely yet and insist in see them as a chaotic, fragmented reality that has nothing to do with us due to the profit-oriented perspective we live with, does not mean that we need to force their dynamic in our classical dramaturgy in order to see them as interconnected beings.

We can make Antipplot or Nonplot films that can express the interconnectedness of our planet by giving the plant kingdom prominence, or rather, completely ceding the spotlight to them. The human language, even if it is the audiovisual one, should transcend the human cause-effect logic, because the language of the plants is not necessarily extravagant, deliberately exaggerate or absurd when we start focusing on feeling, observing and learning. There is interconnectedness in the disconnectedness. That being the case, it is worth mentioning that this is the fertile ground in which dramaturgy beyond the human can operate and flourish.

Wannaz [Wan09] describes the narrative patterns of social realist Arthouse cinema, which plot structure grows organically from the conception of the protagonist and the dramaturgy reflects their attitude towards life and adapts to the character and their possibilities, in opposition to the mainstream cinema where a tried-and-tested plot structure is imposed on the protagonists.

In these alternatives there are characters who do not change their strategy (cyclic situations) or who give up and settle for something that is neither better nor more beautiful than the initial goal, because the characters do not have the power or the means to change their destiny. Sometimes a second character can take more prominence as the protagonist. This produces that the confrontation phase recedes and that the suspense curve stays flat, so that the stasis moments dominate and the character development does not take place at all or only in a very slow manner in contrast to the mainstream films. This further emphasizes the authenticity of the lives depicted and the "Happy end" is not necessarily part of the repertoire as much as a tragic or an open end.

According to Wannaz these alternative ways are able to generate a previously unseen form of empathy with the protagonists. This occurs because both the characters and the public feel powerlessness and frustrated: the former due to the lack of control over the overwhelming circumstances and the latter due to the lack of control over the unforeseeable cinematic events. This stasis, cyclic situations, the impulse to give up that are normally not as appealing as the hero of McKee's Archplot or the hunter in Le Guin's analysis, open a path in which the empathy is the rich soil in which a plant life's dramaturgy can flourish. We strongly recommend exploring, experimenting with, and expanding these alternatives in creative processes involving the plant kingdom. To this end, we suggest reviewing the work of Emmanuele Coccia [Coc19], Stefano Mancuso [Man18], Monica Gagliano et al [Gag17] and Eduardo Kac [Kac07], who have written about plant life with empathy, respect, openness and a desire to learn from these beings.

Providing further insight into our understanding of empathy, which was crucial in establishing this blueprint and is essential for the implementation of this framework, it is important to mention the work of Henri Bergson [Gag60], who considers that while science analyzes and describes things using existing symbols, and de-

depends on the point of view from which it observes, metaphysics uses intuition and empathy with the object of study to go inside it and coincide with that which makes it unique and therefore inexpressible in relation to other objects, such as the duration and essence of the object. Empathy is the path that science, other disciplines like filmmaking, and practical and ancestral knowledge have in common in our framework, because they all use it to know, describe, understand, feel and connect with other beings, which is what we need to co- create with the plant kingdom. In other words, empathy is a postcolonial and ecofeminist practice.

Ultimately, in the first step of our framework posthumanism, ecofeminism, alternative visions of art and dramaturgy together with the empathy and the in-depth knowledge of the plant kingdom are the fundamental elements that are constantly forming and being formed by the artwork that is created in the next steps. Each component of this step broadens the scope of storytelling, providing an ideal environment for co-creating with the plant kingdom. Just as photosynthesis enables both plants and the environment to thrive by exchanging oxygen and carbon dioxide, this step facilitates a similar symbiotic relationship between the artwork and its context.

3.2. Step 2: Recollection

Having identified the contextual environment that enables a creative process involving plant life, this step is inspired by the moment when the ceiba tree takes advantage of the rainy season to absorb the water it needs to produce its own food and to store water for itself and neighboring species for the dry season. We recommend identifying the plant species with which the creators want to work, observing how they behave, and feeling and connecting with them. This process can be enriched by science and other academic disciplines, together with ancestral and practical knowledge by means of the empathy. Just like Le Guin [Gui19] we advise to vindicate the role of the collector and caretaker of the resources by recollecting existing theories, practices and knowledge from different sources that can enrich the desired artwork.

It is our pleasure to affirm that science has also walked in the direction of a different interaction with the plant life beyond the profit. The following examples provide insight into how science, technology and ancestral knowledge enrich the framework. There are scientific examples of ways to help plants with sensors and wireless networks thrive in optimal conditions [Bar06], which could help compensate for the damage caused by our extractive way of inhabiting this planet. There are persons who are using plants to create data clouds [CC20], while others propose living media beings to talk about sociological and environmental topics appealing to the human empathy [FO09]. That very empathy has also been included in scientific approaches where Chinese philosophies [TS16] are the inspiration to defend the plant kingdom from human translation like text, musical tones or voices and emphasizing that the human body and the body of the plant life can be seen as sensors that do not necessarily need electrical components. There is also a project about the ways in which humans can interact with real and artificial plants with an interesting approach about how that interaction can change the way humans can start behaving differently towards nature [PI12]. These and other scientific approaches can

be found in the aforementioned paper by Chang et al. [CM22]. It is important to remark that contrary to what is traditionally said, science increasingly works with abstract concepts such as empathy, emotions, and care, and does not limit itself to the exact sciences. This already provides us with a fertile ground for working with other disciplines.

In this part of the framework all disciplines and sources of knowledge are welcome. The step one can definitely provide some ideas of what and how to collect. However, depending on the topic and plant species involved in each creative process, this step involves observing, selecting and recollecting the particular elements needed for the project. These elements must align with those identified in the contextualization step to ensure a more-than-human approach.

3.3. Step 3: Storage – Transformation

This step is based on the moment when the tree stores water and reserves it until it considers it is time to share it. For us this means the creation of the artwork and its dramaturgy until we are ready to share it. After immersing in the general context and collecting the specific elements of the artwork to be created in the previous steps, the third step is the transformation of the collected elements into the concept of the artwork.

This moment of the process has to do with the organization of the collected elements, the nature of the artwork and the role of the creator. In a world where hours of audiovisual content are being uploaded and watched right now as we, the reader and the author, interact through this paper, it is important that new dramaturgies and forms of artistic expression move beyond the repetition of traditional formulas that merely reproduce the extractive, colonial and chauvinist aspects of our society. Instead, they should harness the power of the media to promote care, connection and vindication.

For the organization of the collected elements we resonate with George Landow [Lan92], who announces a paradigm shift that we consider essential for this step. He proposes replacing the relativity of notions such as center, edge, margin, hierarchy, and linearity with multilinearity, nodes, links, and networks. This idea comes from his analysis of hypertext, an electronic text and an editing mode where writing is non-sequential and branches off. The reader chooses the itinerary, as they can move from one text to another, which is not possible with a printed book.

In other words, there are different multilinear and multisequential paths of verbal and nonverbal information. References in a hypertext can take you with a single click to the exact fragment in the original text from which it was taken, and the reader can decide to read more than just the quote placed in the text they were initially reading. According to Landow, the implication here is that readers are also actively producing a text as they decide the route of their reading rather than simply observing someone else's text. The boundary between reader and writer disappears. In this context, all the constituent nodes of the network are equally important, and it is the reader who charts the route. For this, not only is it necessary for the node system to be complex, but also for the reader to have a hypertextual awareness. Landow also introduces the term "hyper-

media,” which is when nodes with images, videos, sounds, etc. are added to hypertext.

We believe that this approach is beneficial to the creative process because it enables plant life to shape the artwork, blurring the distinction between reader and writer, creator and audience. This openness and flexibility in roles and categories enable classical dramaturgy and art to expand by giving the plant kingdom an active role for a change. Sontag [Son66], for example, has shown that the nature of the artwork can escape the predominant tendency to explain something instead of feeling, which is vital to question the rationalism and anthropocentrism to make way for empathy and care. She questions the predominance given to content over form and to mimesis as the main characteristic of art. Since scientific thought became institutionalized, the approach to art has been based on interpreting works, to the detriment of the works themselves.

This situation has led, according to her, to human sensory capacity and energy being affected, as the intellect has sought to reduce and impoverish works, with the excuse of “explaining” them. What makes works of art stand out, Sontag argues, is not their content, but when they manage to escape interpretation, thanks to their own formal vocabulary and the consequent liberation of the audience from the anxiety of interpreting them, due to their direct message and symbolic liberating quality. This resonates with the poetic message of Eco and the synaesthetic cinema of Youngblood, previously mentioned in the first step of the framework. It is also worth reviewing the work of Brandstetter [Bra24], mentioned in related work, since the use of prepositions can help to think the relationship with the more-than-human in unexpected and creative ways.

After connecting the collected elements and defining the nature of the storytelling and the artwork, it is also important to reflect on the role of the creator in order to complete this step. Youngblood proposes an artist who is an ecologist, that is, a person who seeks to connect and reveal the relationships that exist in the environment between physical and metaphysical phenomena, as opposed to a person who only creates objects, because that vision of art is what makes it the essential instrument for developing human consciousness, humanitarian cooperation, and the relevance of artistic explorations that challenge conventions. This vision can be enriched by the related work of Eeg-Tverbakk [ET23] mentioned above and by some authors like Jacques Lacan [Lac66] and Maria del Rosario Acosta López [AL06], who have written about aesthetic experiences and how one can enrich and transcend human rationality to connect with metaphysical and unknown worlds that do not fit into dominant human categories, like the behavior and intelligence of the plant kingdom.

There is a tendency to think that “putting ourselves in someone else’s shoes” is a practice that can generate empathy, however, in doing so we may be engaging in a self-centered act. Saidiya Hartman [Har97] reminds us the importance of not putting ourselves in the position of another being in order to understand who they are or what they have been through, because we run the risk of eliminating that other being in our selfish quest for self-perception. The context of her work is the representation and analysis of racism and can be extrapolated to the dominant relationship that we humans have with plant life and other detrimental human dynamics of hate and discrimination. Her point of view is similar to the work

of Žeželj [Že22], mentioned above, in the sense that the agency of another being should not be the requirement in order to treat them with dignity and respect.

The fundamental relationship we visualize between the creators towards the plant kingdom and the creative process is a caring relationship. Marta Malo [Mal21] challenges and redefines the care practices in our society. Taking care of others should not be a social or cultural imposition on a specific gender, but rather a political and historical action in which everyone is involved and responsible. It is important to stop caring for those who oppress, violate, kill, abuse and exploit us out of obligation and prioritize caring for beings who want us to be alive and free from debt, and who are willing to take responsibility for our existence alongside others: humans and more-than-humans. Malo proposes care as a rigorous practice of cultivating the bonds we desire, for which we are willing to take risks. We consider that the workflows and technologies used in the projects should be aligned with these views, in order to guarantee care practices.

As has been demonstrated, this step involves taking the general and specific elements from steps one and two, and conceptualizing the story of the artwork in a way that reflects the interconnected nature of humans and non-humans so that the well-being of all parties can be guaranteed. This is possible when nature’s knowledge and human’s knowledge work together without hierarchies, placing empathy and care at the center. This involves entering unknown territory that goes beyond what we can explain in human terms.

3.4. Step 4: Sharing - Building

The final step draws inspiration from the moment when the Ceiba tree shares its stored water with other species during the dry season. In our case this is the moment when the storytelling and the artwork are shared with the world. Since each artwork and dramaturgy is unique, we focus in some tools that can be useful when the concept from the previous step becomes a body interconnected with the plant kingdom and the human realm.

Youngblood invites us to evoke rather than explain. For him, evocation is a poetic form of communication that requires the audience to make a creative effort. They cease to be passive recipients of a pre-established story and become participants in an experience created in the encounter.

To achieve evocation, we recommend the installation and the paracinema. Of course, these are not the only ways to achieve it, but since our background is in audiovisual creation, these are paths that we suggest. Josu Larrañaga [Lar01] defines installation as an experiential art form that blurs the boundary between art and life, allowing the work to be explored, touched, and become a space for networked interaction of different discourses, to which the audience also belongs. This approach preserves the interconnected and horizontal nature that has already been outlined in the previous steps of the framework.

Esperanza Collado [Col12] coined the term “paracinema”. It refers to cinema that is detached from its objectivity in order to question its limits and generate experiences that transcend classic cinematic representation and illusion. For the author, the body and

perception of the audience is fundamental, because they connect film and other arts. Collado defines the materiality of cinema as consisting of two elements. On the one hand, there is objectification and physicality, represented by the machine, the projector, the film and the screen. On the other hand, there is the intelligible, represented by the luminous event, movement, editing, time, duration and spatial-temporal modularity. According to Collado, the latter are cinema's inherent properties with plastic potential that can be renewed and contradicted to restore their power. The author invites us to detach cinema from its objectivity in order to question its limits, giving it other forms and presenting the absence of one or more components. By isolating the fundamental elements of cinema, such as the modulation of light and time and the editing process, from their apparatus, it is possible to generate cinematic experiences that are totally or partially disconnected from the traditional form and physical media of the medium. This does not mean ignoring the technical aspects that have been used narratively to standardize emotions and effects.

The evocation can be achieved by using the installation and the paracinema as means of transmit the more-than-human nature of the dramaturgy and the artwork. An inspiring stance on this issue is that of Jorge La Ferla [LF07], who celebrates the hybridization of media. He believes that maintaining the purity of the familiar is not as narratively or expressively enriching as the cinematic discourse that has become possible since conventional media, video art and experimental cinema have been integrated to create new works. Let's redefine the media, create hybrids and new ways of transmitting that expansive consciousness that expands the human toward the more than human. Let us shake ourselves and our audiences out of the lethargy of the familiar and learn about and care for other species that make our life in this universe possible.

4. Case study of the application of the framework

There is a latin american example that has been analyzed by Pinchot [Pin22] and Gutierrez [Gut24]. It is a theatre play called "Estado vegetal" (Vegetal State) by the Manuela Infante [UMi23] from Chile. That play shows a single actress who gives voice and scenic space to plants while telling the different stories of humans with different relationships with plants. Based on the mentioned authors we reconstructed the 4 steps of the creation of this more-than-human dramaturgy and artwork.

Step 1: The work is defined as a posthumanistic play where plants and humans are just as important for the storytelling. The works of Le Guin and Mancuso, authors that were described in that step, are mentioned by Infante.

Step 2: The particular elements of her work take inspiration from: Rosi Braidotti, Bruno Latour, Jane Bennett, Katherine Behar, Ian Bogost, Claire Colebrook, Manuel DeLanda, Graham Herman, Michael Marder, Timothy Morton y Jean Luc Nancy, which come from philosophy, biology, anthropology and sociology approaches.

Step 3: The nature of the play places the plants in the center and denounces anthropocentrism as a political act. The human and the plant life are seen as one.

Step 4: The play is constructed through a network of voices that

branch out and function modularly, with the aim of making the structure of the work resemble that of the plant world. Only one actress is on stage with plants. The plants have a voice and the human problematize being a plant by being in situations that plants experience and reflecting on that.

5. Discussion

The plant kingdom is in danger because of human action, not because plants lack intelligence or agency. If we measure these aspects from a human point of view, we create a bias that makes us the rulers and center of the world. In doing so, we miss the opportunity to learn from beings that produce their own energy source and that take care of their fellow beings. Of course, there are plant species that can affect others, like eucalyptus trees, which absorb so much water that they cause other trees to die. However, we are focusing on species that take care of themselves and others.

We are already experiencing the drought of the plant kingdom, both literally and figuratively, due to global warming and humanity's exploitative behavior. Therefore, it is a good idea to learn from the Ceibas and prioritize the well-being of the ecosystem, rather than only considering our own benefit. We can become akin to water, which has the capacity to nurture the flora during the arid season and sustain human life in the process.

The entertainment industry and the art institutions have a wide reach and shape our view of the world. However, some of these perspectives follow formulas that have proven to be profitable and generate economic gains, but they do not generate reflection or actions that give the plant kingdom the prominence it deserves. Of course, people who are part of this industry should be able to have decent, well-paid jobs. However, it is important to question the way we create works of art or entertainment and seek to benefit the plant kingdom, without which life would not be possible. It is important to seek a balance between the dynamics of the industry and the well-being of the environment.

Existing interdisciplinary, more-than-human frameworks do not consider dramaturgy or audiovisual expression and are based on subjects such as ghosts or the moon. While some approaches consider language and the dramaturg's role in the creative process, offering opportunities for reflection, they lack connection to traditional or alternative dramaturgies and have no transdisciplinary approach. Our framework enriches and connects these processes because it provides a way of navigating in the vast ocean of theories and practices that surround and are at the heart of the plant kingdom, the dramaturgy and the audiovisual creation and allows us to create stories and works of art with the plant kingdom that generate a collective consciousness in which plant and human life are on the same level.

The creation of a corpus of more-than-human works of art that applies our framework must take into account aspects that have already been pointed out by some of the dramaturgs cited earlier in this paper. McKee [McK97] talks about the way that dramaturgical avantgarde is dynamic and how what today is artistic can be mainstream tomorrow. The nature can and should be included in the way we tell and consume stories. Hopefully one day to take care of the plant kingdom and all the more-than-human beings will

be mainstream and we will use artificial Intelligence to improve our ways of understanding and feeling with plants with the help of academic and ancestral knowledge. However, as pointed out by Wannaz [Wan09], in order to create films that are still for a general public, these new variations of the classic paradigm may need to be combined with epic narrative patterns like the three-act structure so the audience can transit from familiar formulas to expansive artworks. In this regard, we recommend the revision of Aronson's work [Aro10] where a detailed description of the different types of parallel narratives, including linear and non-linear is offered, which according to him depend on the content and the philosophy of the film and are all based in the three act structure. It is important to mention that his approach is based on films that can be categorized in McKee's work as Archplot or Miniplot, so it can be taken into account in the process of building a bridge between plant kingdom dramaturgy and human dramaturgy.

6. Conclusion

The plant kingdom is not a public to impress, but a being that is interconnected and has a lot to share with us and the world. It deserves a better story than simply being an object that we use and harm, because it is a generous life creator that never stops producing the oxygen we use to live.

The Ceiba tree not only does that, but also absorbs, stores and shares water for the dry seasons for the well-being of the ecosystem, of which it is a part. This makes it a caretaker with a radically generous approach to life, which is very different from the profit-based mentality of human society. The more-than-human co-creation of stories and artworks with the plant kingdom is born in the caring practice of the beings that make life. The Ceiba is a natural artist whose story is one of self-improvement and that of its fellow beings, even in the most difficult times. It is also the story of the responsible use of resources in a spirit of solidarity.

The plant kingdom is in a constant creative and communicative process that is no less important than that of humans. Dramaturgy is an essential component of human storytelling and artistic creation, but its anthropocentric approach can be challenged and expanded through posthumanism, ecofeminism, empathy, care practices and the exploration and hybridization of artistic forms. This allows us to create ways in which plants can be represented as active sentient beings that can create stories in a more-than-human way from which we can learn and expand the limited human-centered perspective we live in. To give the plant life room in the creation of the storytelling is a revolutionary step that reinforces existing initiatives that are compromised with the dignity and equality of all beings.

Our framework is an adaptable blueprint of 4 parts that can be implemented by persons interested in co-creating with the plant kingdom, especially audiovisual creators who want to create alternative storytelling possibilities based on a transdisciplinary approach. The steps are designed based on the biological processes of the ceiba tree and provide insight into how the contextual and particular elements of creative processes can be organized. They also provide information about the nature of the artwork, the role of the creators, and suggestions for how hybridization, the resignification

of audiovisual aspects, and installation approaches can help create works that bring the creative processes of humans and the plant kingdom together.

There are already scientific approaches that combine qualitative and quantitative research and we can take advantage of this to strengthen the connections between science, art, philosophy, technology and ancestral knowledge. All disciplines are telling a story, and the union of all these diverse stories is the history of humanity. If there is also room for the plant kingdom in this intrinsic creative process, we are not only expanding our human knowledge, but also building on difference, empathy and care, which can bring answers to questions that can only be answered from a transdisciplinary complexity, such as the measures that must be taken to curb climate change, social inequality, and the way in which plant beings can also teach us to live in balance and independence, as they do. The way we tell stories are a reflection of the perspectives we have of our world, but also the building blocks of the future world we are creating in the present. We can create a world where plant kingdom and human kingdom are not separated anymore, where wealth is produced not only in economic terms but also in terms of care, empathy, dignity and creativity.

7. Future work

The focus of our framework is the plant kingdom, while a more-than-human dramaturgy with animals and fungi have been explored by other authors. Even though it is a complex task, we consider important to integrate those explorations in order to expand the scope and the interconnection of the more-than-human dramaturgies and artworks.

It is imperative to keep on connecting the way different disciplines work on the more-than-human world and to continue creating artistic processes like the ones we saw on the paper in order to enrich the plurality and the network of the way humans and plant life get together.

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